



Machine Learning

Instructor led Training and e-Learning

What is this course about?

This course provides a structured understanding of machine learning models, covering supervised, unsupervised, and ensemble techniques used across industries today. Learners will explore model selection, training, evaluation, and optimization, gaining hands-on exposure to widely used algorithms such as regression, classification, clustering, decision trees, and neural networks. By the end of this course, participants will be equipped to build, analyze, and deploy ML models confidently.

Key Features

1. Duration

Live-online: 2 Days

2. Level: Intermediate

3. Language: English

4. Prerequisites:

- Basic understanding of Python programming.
- Familiarity with statistics and probability.
- Fundamental knowledge of linear algebra concepts.
- Prior exposure to data analysis is helpful but not mandatory.

- Ability to work with datasets and tools like Jupyter Notebook.

Learning Path:

- Introduction to Machine Learning Foundations.
- Supervised Learning Models (Regression & Classification).
- Unsupervised Learning Models (Clustering & Dimensionality Reduction).
- Ensemble Methods & Model Optimization Techniques.
- Model Evaluation, Deployment, and Real-World Use Cases.

Related Courses:

- Introduction to Data Science
- Python for Data Analysis
- Applied Machine Learning
- Deep Learning Fundamentals.

Certification:

- Microsoft Azure Data Scientist Associate (DP-100)
- Google Professional Machine Learning Engineer Certification

5. **Delivery:** Live-online, Classroom, Self-paced, Blended

What are the objectives of Machine Learning Models training?

At the end of this training, you will be able to:

- Understand the core concepts and types of machine learning models.
- Learn to build and evaluate supervised and unsupervised models.
- Apply feature engineering and model optimization techniques.
- Gain hands-on experience with real datasets and ML tools.
- Develop skills to select the right model for specific business problems.

Who is this Training for?

- Data analysts looking to advance into machine learning.
- Software developers seeking ML model-building skills.
- Students or graduates interested in AI/ML careers.
- Business professionals wanting to understand ML applications.
- Anyone aspiring to work in data science or predictive analytics.

Course Outline:

- Introduction to Data Mining and Machine Learning – Various Use Cases
- Applied Machine Learning vs. Machine Learning Research
- Data Acquisition
- Exploratory Data Analysis (EDA)
- Relationship Identification in Data
- Techniques for Data Munging
- New Variable Creation from Existing Data
- Supplementing Data from Allied Sources
- Supervised vs. Unsupervised Learning Techniques
 - Regression
 - Classification
 - Clustering
- Matching Appropriate Algorithms with Input Data
- Test and Cross-Validation Datasets
- Concepts of Overfitting and Underfitting
- Regularization Techniques
- Techniques to Handle Overfitting and Underfitting
- Managing and Understanding Input Data
 - Numeric Data
 - Nominal Data
- Performance Metrics and Model Evaluation
- Concepts of Weak and Strong Models
- Discussion on Various Machine Learning Algorithms and Their Applications
 - Decision Trees
 - RIPPER
 - Apriori
 - Support Vector Machines (SVM)
 - Naïve Bayes
 - K-Means
 - K-Nearest Neighbours (KNN)
- Case Studies and Practical Implementation of the Above-Mentioned Algorithms Using Kaggle Datasets